

TECHNICAL SPECIFICATION MANUAL

EARSAFE METER V1

Battery-powered desk sound monitor. It has an e-ink display. It tracks daily sound dose. It gives clear warning states during long listening sessions.

System Intent

- Measure room sound near the user position.
- Show present dBA, short-term average, peak, and daily dose.
- Give a clear warning before sound exposure becomes excessive.
- Operate on battery power with USB-C charge.
- Use common modules for a first build.

DOCUMENT DATA

Project: EARSAFE METER

Revision: A

Status: V1 specification

Owner: builderfriday

Audience: builder and non-specialist user

TARGET USE: DESK MONITOR

LOW POWER E-INK UI

CALIBRATION REQUIRED

SECTION 01 Scope

This document gives the technical definition for EARSAFE METER V1. The device is a small desk unit. It measures sound at the listener area. It shows the sound condition on an e-ink display. It tracks daily exposure dose. It gives a warning when the sound level is too high for too much time.

The device is for personal control and awareness. It is not a certified legal instrument. Use calibration to make the reading useful. Do not use the device as the only control for industrial safety or legal compliance.

IMPORTANT

The specification uses short sentences and direct commands. The goal is clarity. The goal is not hardware jargon.

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SECTION 02 **System Summary**

PRIMARY FUNCTION

Measure A-weighted sound level at the desk and convert it to a clear user display.

MAIN USER NEED

Know when the room is too loud before daily sound dose becomes high.

V1 DESIGN RULE

Use simple modules. Keep wiring low. Keep battery life practical.

ITEM	V1 TARGET	REASON
Form	Desk meter	Easy to place near the listening position.
Display	2.13 in e-ink	Low power. High readability. Professional look.
Controller	Nordic nRF52840 class board	Good low-power performance. Enough GPIO. BLE support.
Microphone	Digital MEMS microphone	Stable digital path. Simple integration.
Battery	3000 mAh LiPo nominal	Useful run time without large size.
Charge Interface	USB-C	Simple user charge path.
Core Metrics	Current dBA, 1 min average, peak, daily dose	These values tell the user what to do next.

SECTION 03 Top-Level Requirements

USER REQUIREMENTS

- The device must stand on a desk without a separate mount.
- The display must be readable from about 0.5 m to 1.5 m.
- The user must know the sound state in less than 2 seconds.
- The device must charge with a standard USB-C cable.
- The device must keep operation simple. V1 will not depend on Wi-Fi.

SYSTEM REQUIREMENTS

- The signal path must support A-weighted level calculation.
- The firmware must compute 1 second sound updates.
- The firmware must compute daily sound dose with a 3 dB exchange rule.
- The device must store calibration offset in non-volatile memory.
- The device must show a battery-low state before shutdown.

ID	REQUIREMENT	PASS CONDITION
REQ-01	The device must show current A-weighted sound level in dBA.	User can read the value on the main screen.
REQ-02	The device must show warning state at 80 dBA and alert state at 85 dBA.	Screen state changes at the configured thresholds.
REQ-03	The device must track cumulative daily sound dose.	Dose value changes during a controlled sound test.
REQ-04	The device must operate from internal battery power.	Device runs without USB cable attached.
REQ-05	The device must support one-time calibration with a reference meter.	User can store offset and keep it after power cycle.

SECTION 04 System Architecture

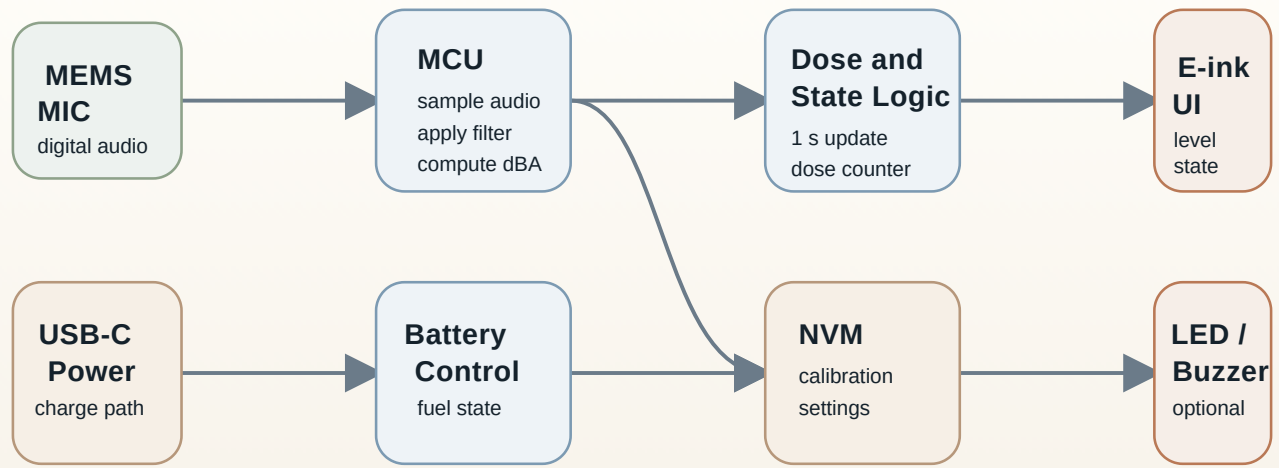


Figure 1. Signal path and control path. The sound enters through the digital microphone. The MCU calculates the sound level. The logic then updates the display and stored data.

SECTION 05 Hardware Units

UNIT	PREFERRED PART CLASS	FUNCTION	NOTES
Main controller board	XIAO nRF52840 Sense or equivalent	Run firmware, low-power control, BLE service if enabled later	Pick a board with LiPo charge support if possible.
Microphone	On-board digital mic or external I2S/PDM MEMS mic	Capture stable digital audio	External mic can give better placement freedom.
Display	2.13 in black/white e-ink module	Show large values with low power use	Avoid tri-color for V1. Refresh is slow.
Battery	1-cell LiPo, 3000 mAh nominal	Give all-day portable use	Capacity can change after power test.
Charge interface	USB-C 5 V input	Charge the battery and allow cable operation	Use protected battery parts only.
User input	1 to 2 push buttons	Change screen, start calibration, set threshold	Keep the UI simple.
Optional alert	Low-current LED or piezo buzzer	Give fast warning without full display refresh	Can be removed in first quiet prototype.

PREFERRED V1 HARDWARE DIRECTION

- Use a development board first.
- Do not design a custom PCB in V1.
- Use module headers or short wire runs.
- Keep the microphone near an acoustic port in the case.

ITEMS TO AVOID IN V1

- Always-on Wi-Fi.
- Complex touch interface.
- High-refresh color display.
- Large amplifier or speaker parts inside the case.

SECTION 06

Firmware Functions

SIGNAL PROCESSING

- Read digital microphone samples.
- Remove DC bias if needed.
- Apply A-weighting filter.
- Compute RMS energy over the update window.
- Convert to dBA with stored calibration offset.

EXPOSURE LOGIC

- Calculate 1 second level updates.
- Track 1 minute average level.
- Track fast peak level.
- Track daily dose with 3 dB exchange rule.
- Reset daily dose at user-set day boundary.

STATE	ENTRY CONDITION	USER DISPLAY
Normal	Below warning threshold	Large level value and calm status mark
Warning	At or above 80 dBA for active window	Strong caution mark and higher visual contrast
Alert	At or above 85 dBA for active window	Clear alert mark with optional pulse indicator
High Alert	At or above 94 dBA	Immediate risk state. Optional buzzer allowed.
Low Battery	Battery under low limit	Battery warning and reduced refresh rate

SECTION 07

User Interface

MAIN SCREEN

Show current dBA in large type. This is the first value the user must see.

SUPPORT DATA

Show 1 minute average, peak, and dose in smaller type.

FAST MEANING

Use simple words: Normal, Warning, Alert, High Alert.

UI ELEMENT	TARGET STYLE	PURPOSE
Large level field	At least 28 pt equivalent on screen	Read the sound level from across the desk.
Status field	One short word	Give a decision without number analysis.
Dose field	Percent of daily target	Show long-term exposure progress.
Battery field	Small icon or percent	Prevent surprise shutdown.
Calibration mode	Temporary setup screen	Store reference offset.

DISPLAY RULE

The display must not look busy. One number must lead the screen. The user must know the condition with a quick look.

SECTION 08

Power and Battery

POWER CONCEPT

- Primary mode is battery operation.
- USB-C is for charge and optional desk power.
- Display refresh must be event-based, not continuous.
- Wireless functions must stay off in the default mode.

BATTERY STRATEGY

- Use a protected LiPo pack.
- Warn the user before low-voltage shutdown.
- Save settings before deep discharge.
- Test real run time with display and microphone active.

PARAMETER	V1 TARGET	COMMENT
Nominal battery	3000 mAh	Good size for a desk unit.
Charge input	5 V USB-C	Do not use higher-voltage PD in V1.
Expected run time	One full day target	Real value depends on display and mic duty cycle.
Low battery warning	20 percent remaining target	Threshold can change after test.

SECTION 09 Calibration and Accuracy

The microphone gives digital sound data. It does not give true dBA by itself. The system needs calibration against a reference. For V1, use a known sound meter at the same position and the same test sound. Save the difference as the device offset.

CALIBRATION METHOD

- Put the reference meter and the device at the same point.
- Use steady pink noise or other stable test sound.
- Wait for level stability.
- Read both values.
- Store the difference as offset.

ACCURACY STATEMENT

- Before calibration, use values as relative guidance only.
- After calibration, use values as personal safety guidance.
- Do not claim legal meter accuracy.
- Repeat calibration after major hardware change.

KEY RISK

A professional display does not make an uncalibrated reading true. Calibration is the line between a nice gadget and a useful sound guide.

SECTION 10

Mechanical Design

FEATURE	REQUIREMENT	DESIGN NOTE
Form factor	Self-standing wedge or upright desk body	The front angle must support easy reading.
Microphone opening	Direct acoustic port to room	Do not block the port with foam or dense wall sections.
Display window	Flush or recessed front face	Keep reflections low.
USB-C access	Easy cable insert from side or rear	Do not make the cable bend sharply.
Battery enclosure	Protected internal bay	Keep battery away from sharp edges.

CASE BUILD DIRECTION

- Use a two-part 3D printed shell.
- Use screw posts or captive inserts if needed.
- Keep the interior layout modular.
- Leave service access for the battery and display cable.

FINISH DIRECTION

- Keep outer geometry quiet and clean.
- Use clear labels only where needed.
- Avoid decorative cutouts near the microphone.
- Make the front face feel like a real instrument.

SECTION 11 **Build and Verification**

PROTOTYPE BUILD ORDER

- 1. Make the electronics work on the bench.
- 2. Show a basic e-ink screen.
- 3. Read live sound data.
- 4. Compute dBA and dose.
- 5. Calibrate with a reference meter.
- 6. Move the system into the 3D printed case.

VERIFICATION TESTS

- 1. Power-up and charge test.
- 2. Battery-only run test.
- 3. Display readability test.
- 4. Threshold change test at 80 and 85 dBA.
- 5. Dose accumulation test.
- 6. Calibration retention test after reboot.

TEST ID	TEST	EXPECTED RESULT
T-01	USB-C charge test	Battery charges and system remains stable.
T-02	Sound response test	Level changes with source volume change.
T-03	Threshold state test	UI changes at configured limits.
T-04	Daily dose test	Dose value increases with controlled exposure time.
T-05	Battery-life test	Measured run time meets or informs the target value.

SECTION 12 Risks and Controls		
RISK	EFFECT	CONTROL
Uncalibrated level	User trusts a wrong value.	Require calibration step and show status in setup.
Poor microphone placement	Reading does not represent user position.	Place the port on the face that points to room sound.
Short battery life	Device becomes inconvenient.	Use e-ink, low-power firmware, and event-based refresh.
Over-complex UI	User ignores the device.	Keep one large number and one clear state word.
Unsafe battery pack	Damage or heat event.	Use protected cells and proper charge hardware only.
Revision A		
Prepared for builderfriday		